

LLM-MTPSO: Large Language Model-Assisted Multi-task Particle Swarm Optimization for Imbalanced Classification

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Abstract—Imbalanced data widely exists in security-prioritized domains, where minority classes often bring a disproportionately high risk. Especially in multi-class scenarios, challenges arise not only from the imbalance between majority and minority classes, but also from the difficulty in separating minority classes and their underrepresentation in the training data. To address these issues, this paper adopts a One-Versus-One decomposition strategy to transform the multi-class problem into multiple binary subtasks. This formulation naturally increases the visibility of minority classes. Furthermore, we propose a large language model-assisted multi-task particle swarm optimization (LLM-MTPSO) algorithm. Each subtask is independently optimized via PSO, while a LLM is leveraged to semantically fuse optimal solutions among similar subtasks. This mechanism facilitates knowledge transfer and refinement at the solution level, enhancing the model's capacity to recognize minority patterns. Experiments on nine imbalanced datasets show that the test results of our method in terms of imbalanced classification accuracy outperform four classical methods. The ablation experiments indicate that the introduction of LLM promotes information sharing among subtasks, thus improving the algorithm performance.

Index Terms—imbalanced classification, large language model, multi-task learning, particle swarm optimization

I. INTRODUCTION

In the domain of industrial fault diagnosis, medical analysis, and financial fraud detection, imbalanced data is prevalent. As more types of data are continuously increasing in volume, identifying the minority classes in massive imbalance datasets has become increasingly challenging. If at least one class in the dataset represents significantly fewer instances than other classes, the dataset is classified as imbalanced [1]. In real-world industrial scenarios, machines typically remain in normal operating conditions, resulting in a significant surplus of healthy samples compared to fault samples [2]. However,

despite the extremely low proportion of fault samples, they often exist diverse failure modes (e.g., bearing wear, shaft misalignment, rotor imbalance). These abnormal vibration signals with diverse types may indicate equipment failures and directly affect production safety. Therefore, although the number of minority class samples is small, they carry the core risks of system safety. Their importance is not proportional to the quantity but directly related to the consequences. Considering imbalanced data classification scenarios, several researchers have made numerous innovative studies. These studies can roughly be divided into two categories [3]: the first type is data-level improvement and the second type is algorithm-level optimization.

Improvements at the data-level mainly focus on sampling, including upsampling and downsampling, to rebalance the number of classes in the original dataset. Oversampling (such as SMOTE and its variants [4], [5]) alleviates the imbalance problem by synthesizing minority class samples, while undersampling (such as ensemble-based methods like EasyEnsemble [6]) strategically reduces majority class data to retain key information. However, these methods have inherent limitations: oversampling risks amplifying the noise of the minority class, and undersampling may discard valuable majority-class patterns. Especially in complex and imbalanced scenarios, challenges in generalization and distribution distortion still exist. At the algorithm level, many methods attempt to adapt standard classifiers to better handle class imbalance. Cost-sensitive learning (CSL) is a classical approach that incorporates manually defined cost matrices to penalize misclassification of minority classes [7]. However, manual design is often difficult to generalize and may introduce human bias. In contrast, evolutionary algorithms (EAs), which simulate natural selection and genetic variation, have gained attention

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for their ability to perform global optimization without requiring gradient information or explicit priors [8]–[10]. Recent studies have explored EAs in various forms, such as multi-objective genetic programming [11], set-based feature selection (FS) [12], and particle swarm optimization (PSO) variants with customized operators [13], [14]. Despite their success, most traditional EAs treat all instances equally, making them prone to favoring the majority class and overlooking minority-specific patterns. This often leads to biased classifiers with reduced generalization in extremely imbalanced scenarios [2].

Since 2022, the rise of large language models (LLMs) has brought new vitality to various fields, offering powerful capabilities in semantic modeling and contextual reasoning. To overcome the limitations of traditional EAs in handling imbalanced classification, we propose a large language model-assisted multi-task PSO (LLM-MTPSO) framework. By decomposing the original task into multiple One-Versus-One (OVO) subtasks, the model allows for finer-grained learning of minority class distinctions and reduces optimization bias. Furthermore, LLMs are leveraged to semantically fuse optimal solutions from similar tasks, enhancing the overall knowledge integration at the solution level. This synergy between task decomposition and LLM-guided fusion enables the algorithm to better capture minority-relevant patterns. Experimental results on nine imbalanced datasets validate the effectiveness of the proposed method compared with several representative baselines.

II. RELATED WORK

In recent years, Multi-Task Learning (MTL) has gradually demonstrated advantages in imbalanced classification problems. MTL splits complex original tasks into multiple sub-tasks, such as the OVO or One-Versus-Rest (OVR) structure, enhancing the local exposure of minority class samples to alleviate biases dominated by majority classes [15]. In the field of high-dimensional FS, EAs are widely used for sub-task optimization due to their gradient-free property and global search capability. For example, MEL [16], PSO-based multi-task evolutionary learning significantly improves FS performance through independent particle swarms and information sharing mechanisms. Although the combination of MTL and EAs has made theoretical and experimental breakthroughs, it still has critical limitations [17]. When sub-tasks have severe class imbalance, EAs often optimize for overall performance, ignoring the personalized optimization needs of minority classes. This may lead to negative transfer or majority class bias, affecting model generalization.

Recent studies have explored introducing semantic or knowledge-aware mechanisms into EAs to enhance the refined performance of migration strategies. For instance, some works introduce gradient coordination mechanisms like PCGrad [18] to reduce conflicts between sub-tasks, but these methods primarily focus on numerical optimization rather than semantic levels [19]. In the domain of EAs, researchers have proposed population structure-aware strategies like co-evolution and

multi-factor evolution [20], yet these still rely on explicitly predefined task interactions and lack automatic semantic alignment mechanisms. Meanwhile, LLMs have demonstrated exceptional capabilities in semantic understanding and context modeling. Recent attempts have combined LLMs with EAs, such as LMPSO [21] introducing prompt-based semantic update mechanisms in PSO, and using LLMs for automatic generation of evolutionary operators [22]–[24]. Additionally, embedding LLMs into PSO processes by replacing poorly performing particle positions with LLM suggestions can effectively reduce model evaluation costs [25].

However, a key limitation persists in existing methods. Although MTL facilitates problem decomposition and knowledge transfer, the interaction between sub-tasks still relies heavily on manually crafted structures, which often overlook latent inter-task correlations. This leads to rigid and frequently suboptimal knowledge sharing. To overcome this, we introduce LLMs as semantic bridges that fuse solutions from similar tasks within the joint MTL and EA framework. By modeling semantic affinities, LLMs guide context-aware information flow and can adaptively emphasize minority-class learning. Unlike existing studies that primarily focus on prompt tuning or single-task enhancements, our approach enables both independent sub-task optimization and selective cross-task integration. This represents a novel direction, as no prior work has systematically explored sub-task level semantic fusion for imbalanced classification.

III. PROPOSED METHOD

As shown in Fig. 1(a), samples from different classes in the original dataset are paired to form multiple binary sub-datasets. Each sub-dataset corresponding to an independent sub-task focuses on a unique class pair. During the sub-task optimization phase, the PSO algorithm is employed to search for optimal solution in each sub-task’s classification model. By dynamically adjusting particle positions and velocities, PSO efficiently explores potential distribution patterns of minority class samples in sub-feature spaces. Additionally, those involving overlapping classes are grouped into similar task sets. Within each set, sub-tasks form a knowledge-sharing network as neighbor nodes. In the iterative optimization process, sub-tasks can dynamically utilize the historical results of adjacent modules, and integrate them into new feature information through the LLM, thus expanding the solution space. Finally, a soft voting strategy is adopted to probabilistically weight and fuse predictions from all sub-tasks. The above process is as shown in Algorithm 1. This design enables semantic-driven collaboration across sub-tasks while preserving task-specific optimization. The LLM acts as a mediator that selectively propagates informative knowledge, especially benefiting minority-class discrimination.

A. Binary Multi-task PSO

In our proposed framework, the original dataset is decomposed into a set of OVO binary sub-tasks, each independently optimized by PSO with a dedicated particle population. This

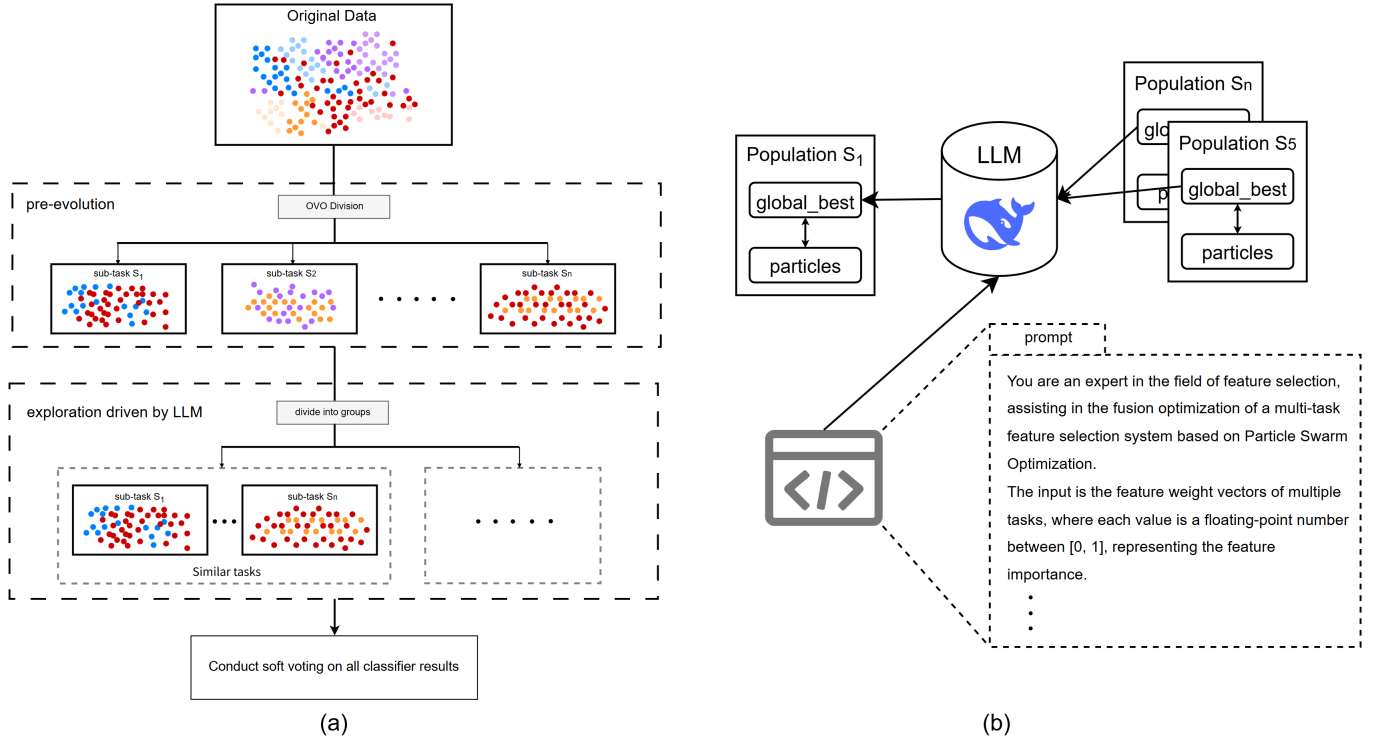


Fig. 1. Points of different colors represent data of different categories. (a) the way of data partitioning in the algorithm of this paper. (b) an example of using a LLM for co-evolution in a set of similar tasks. Assume that the current evolving population is S_1 , and its similar populations are S_5 and S_n .

Algorithm 1 LLM-MTPSO Framework

- 1: Configure algorithm parameters
- 2: Decompose data into OVO binary classification tasks
- 3: **for** each task in task_list **do**
- 4: Perform preliminary optimization for the current task
- 5: **end for**
- 6: **for** each task in task_list **do**
- 7: Identify similar tasks for knowledge transfer
- 8: Explore the solution space guided by a LLM
- 9: **end for**
- 10: Aggregate final classification results via soft voting

decomposition enhances the visibility of minority classes in pairwise comparisons by explicitly separating class pairs. Each particle is represented by a real-valued position vector $\mathbf{X}_i^{(t)} \in \mathbb{R}^d$ and a velocity vector $\mathbf{V}_i^{(t)} \in \mathbb{R}^d$, where d denotes the solution space dimensionality. Its evolution is guided by the personal best $\mathbf{P}_i^{(t)}$ and the task-specific global best $\mathbf{G}^{(t)}$. The update rules for the i -th particle at iteration $t + 1$ are:

$$\mathbf{V}_i^{(t+1)} = w \cdot \mathbf{V}_i^{(t)} + c_1 \cdot r_1 \cdot (\mathbf{P}_i^{(t)} - \mathbf{X}_i^{(t)}) + c_2 \cdot r_2 \cdot (\mathbf{G}^{(t)} - \mathbf{X}_i^{(t)}) \quad (1)$$

$$\mathbf{X}_i^{(t+1)} = \mathbf{X}_i^{(t)} + \mathbf{V}_i^{(t+1)}, \quad (2)$$

where w , c_1 , c_2 are hyperparameters controlling inertia and acceleration; r_1 , $r_2 \sim \mathcal{U}(0, 1)$ are random values drawn from a uniform distribution [26].

When multiple sub-tasks share overlapping samples, they are grouped as neighboring tasks. For knowledge transfer, we introduce a crossover operation between the global best of the current task $\mathbf{G}^{(t)}$ and that of a randomly selected neighbor $\mathbf{G}_{\text{neighbor}}^{(t)}$. This strategy allows the algorithm to incorporate complementary information across tasks without requiring full model retraining. By leveraging the structural similarity among neighboring tasks, the crossover not only facilitates knowledge reuse but also encourages adaptive exploration in underrepresented decision regions. The updated global best $\hat{\mathbf{G}}^{(t)}$ is calculated as:

$$\hat{\mathbf{G}}^{(t)} = (1 - cr) \cdot \mathbf{G}^{(t)} + cr \cdot \mathbf{G}_{\text{neighbor}}^{(t)}, \quad (3)$$

where $cr \in [0, 1]$ is the crossover rate. If the new position $\hat{\mathbf{G}}^{(t)}$ yields better fitness than the current $\mathbf{G}^{(t)}$, it will replace the current global best and be propagated to all particles for subsequent updates. This mechanism enhances swarm diversity and mitigates the risk of premature convergence.

B. LLM-assisted Exploration of Diversity

To enhance the search diversity and facilitate cross-task knowledge transfer, we introduce a LLM as a knowledge explorer that drives co-evolution among similar tasks. As illustrated in Fig. 1(b), the LLM acts as an intelligent fusion engine that collects global best solutions from multiple populations and provides guidance for generating new exploratory solutions. Unlike conventional strategies that rely solely on stochastic variation or fixed fusion rules, LLM can adaptively capture latent semantic associations between tasks, offering

more context-aware and informed search directions. This semantic reasoning capability helps overcome the limitations of manually designed transfer operators.

During the evolutionary process, sub-tasks that share overlapping samples are grouped into similar task sets (e.g., S_1 , S_5 , and S_n). At each iteration, the global best positions from the current population and its similar neighbors are fed into the LLM in the form of real-valued vectors. Each vector represents the feature weights of a solution, where each element lies in $[0, 1]$, indicating the importance of a corresponding feature. This structure allows the LLM to operate on a unified semantic space, making it feasible to generalize across different task sets. As shown in Algorithm 2, a prompt is constructed to instruct the LLM to act as a FS expert, assisting in fusing knowledge from multiple related tasks. The input consists of global best position vectors from the current population and its neighbors, and the LLM outputs a new exploratory candidate $\hat{G}^{(t)}$. The position $\hat{G}^{(t)}$ is evaluated by the fitness function $f(\cdot)$ and compared with the current global best $G^{(t)}$:

$$G^{(t)} = \begin{cases} \hat{G}^{(t)}, & \text{if } f(\hat{G}^{(t)}) < f(G^{(t)}), \\ G^{(t)}, & \text{otherwise.} \end{cases} \quad (4)$$

If the new candidate provides a better fitness, it replaces the current global best and is propagated to all particles to guide their updates in the next iteration. In addition, the invocation of LLM is not unlimited. We stipulate that it is invoked once every $f = 10$ iterations, and for other iterations, formula 3 is used for random crossover to explore diversity. This hybrid design balances the computational cost of LLM calls with the benefits of semantic-level knowledge fusion, ensuring both efficiency and effectiveness during the optimization process.

Algorithm 2 LLM-assisted Fusion

Require: Iteration index t , fusion frequency f , LLM call probability p , current global best $G^{(t)}$, group global bests $\mathcal{G}$, weighting factor α

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1: if  $f > 0$  and  $t \bmod f = 0$  then
2:   if  $\mathcal{G} \neq \emptyset$  and  $\text{rand}() < p$  then
3:     Collect global best vectors from  $\mathcal{G}$  and current  $G^{(t)}$ 
4:     Construct prompt and invoke LLM
5:     Generate binary mask using a random threshold
6:     Evaluate fitness of the fused solution
7:     if fused fitness improves over current  $G^{(t)}$  then
8:       Update  $G^{(t)}$  in a crossover way
9:       Apply random perturbation and clip values
10:      Update current global best and its fitness value
11:    end if
12:  end if
13: end if

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This LLM-assisted mechanism introduces semantic-level diversity into the swarm, allowing knowledge to flow across tasks while respecting the problem structure. As a result, the optimization process not only preserves the local convergence behavior of each sub-population but also gains

enhanced global exploration ability through semantically informed knowledge fusion.

IV. EXPERIMENT AND ANALYSIS

A. Parameter Settings and Datasets

The relevant parameters for algorithm design are summarized in Table I. Among them, the cognitive and social acceleration coefficients (c_1 and c_2) are both set to 1.49445, following the constriction coefficient configuration recommended by [27]. For LLM-based fusion, the crossover rate cr is randomly initialized in $(0, 1)$ and adaptively selected from $[0.6, 0.9]$ to promote meaningful knowledge transfer.

The datasets used in the experiment is selected from [28]. The dataset consists of two parts. One part is the TE benchmark, and the other part is the industrial dataset in the real world. The dataset information is summarized in Table II. The industrial dataset is derived from a bearing test rig that simulates actual working conditions, including vibration data of 4 fault types under 4 working conditions (2 loads $\times$ 2 speeds). Sampled at 25.6 kHz, it undergoes extraction of 33-dimensional features and is divided into a 200-sample imbalanced training set and the remaining test set.

B. Compared Algorithms and Evaluation Metrics

We compared with four imbalanced ensemble algorithms on the industrial dataset and the TE dataset. These algorithms include Self-paced Ensemble [29], BalanceCascade [6], SMOTEBoost [30], and AsymBoost [31]. Among them, Self-paced Ensemble is an ensemble method based on the self-paced learning strategy. It effectively addresses the data imbalance problem by gradually screening samples and dynamically adjusting the ensemble model. Similar to BalanceCascade, it mainly balances the data distribution through sample under-sampling. On the other hand, SMOTEBoost increases the number of minority class samples through sample oversampling, and AsymBoost adjusts the model's learning process by assigning different weights to samples of different classes. Compared with these techniques, our method emphasizes multi-task evolutionary optimization, enabling sub-task specialization and richer representation learning for minority classes. In addition, the integration of LLM facilitates solution-level knowledge fusion across related tasks, complementing local learning with semantic-aware guidance.

In this study, the optimization objective of our proposed algorithm is defined as the minimization of $1 - G\text{-mean}$, which reflects the trade-off between sensitivity and specificity and is particularly suitable for evaluating performance on imbalanced classification tasks. Additionally, to evaluate the classification performance, we adopt Balanced Accuracy (BAC) as the evaluation metric. BAC is widely used in imbalanced learning scenarios, as it provides a fair evaluation across classes by equally weighting sensitivity and specificity. It is defined as follows:

$$\text{BAC} = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right) \quad (5)$$

TABLE I
ALGORITHM CONFIGURATION PARAMETERS

Parameter	Description
rm_p	Random mating probability for cross-task knowledge transfer, set to 0.6
T	Total number of iterations (default: 100)
cr	Crossover rate for LLM-based fusion, randomly initialized in $(0, 1)$, later selected from $[0.6, 0.9]$
d	Number of features in the dataset
N	Number of particles in the swarm, set to $d // 3$
c_1	Cognitive coefficient, set to 1.49445
c_2	Social coefficient, set to 1.49445
w	Inertia weight, become smaller as the number of iterations increases

TABLE II
DETAILS OF DATASETS USED IN THE EXPERIMENT

Dataset number	Dataset type	Imbalanced ratio	Conditions of manufacture
1	industry	10	1200rpm 0N
2	industry	10	1200rpm 3000N
3	industry	10	600rpm 0N
4	industry	10	600rpm 3000N
5,6,7,8,9	TE	10,20,30,40,50	—

where TP , TN , FP , and FN denote the numbers of true positives, true negatives, false positives, and false negatives, respectively.

C. Results and Analysis

As shown in Table III, we compared the average test performance of these methods. The optimal BAC obtained on each dataset will be shown in bold. Significance testing is represented by the column ST. The column ST represents the statistical Wilcoxon significance test results of the corresponding method relative to LLM-MTPSO (significance level is 5%). "+" or "-" indicates that the result is significantly better or worse than LLM-MTPSO, and " $\approx$ " indicates that their performance is similar.

The results clearly demonstrate the superior performance of LLM-MTPSO in most cases. Specifically, LLM-MTPSO achieves the highest BAC scores on seven datasets. In datasets

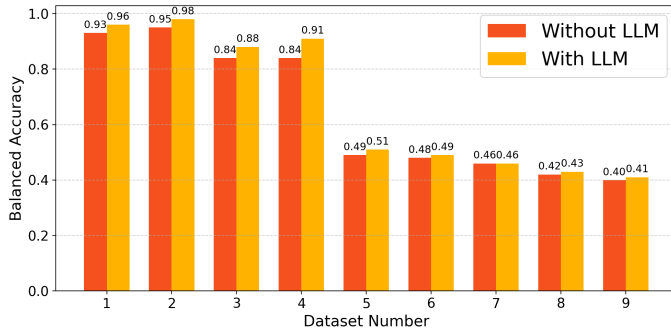


Fig. 2. Ablation study on the impact of LLM. The results show that incorporating the LLM module improves the BAC across most datasets.

TABLE III
BAC PERFORMANCE COMPARISON ACROSS DATASETS.

Dataset number	Method	Mean(std)	ST
1	SelfpacedEnsemble	0.80(0.01)	—
	BalanceCascade	0.80(0.01)	—
	SMOTEBoost	0.91(0.01)	—
	AsymBoost	0.90(0.01)	—
	LLM-MTPSO	0.96(0.01)	—
2	SelfpacedEnsemble	0.77(0.02)	—
	BalanceCascade	0.71(0.02)	—
	SMOTEBoost	0.97(0.01)	$\approx$
	AsymBoost	0.97(0.01)	$\approx$
	LLM-MTPSO	0.98(0.01)	—
3	SelfpacedEnsemble	0.72(0.01)	—
	BalanceCascade	0.72(0.01)	—
	SMOTEBoost	0.87(0.01)	—
	AsymBoost	0.85(0.01)	—
	LLM-MTPSO	0.88(0.01)	—
4	SelfpacedEnsemble	0.82(0.02)	—
	BalanceCascade	0.81(0.01)	—
	SMOTEBoost	0.90(0.01)	$\approx$
	AsymBoost	0.90(0.01)	$\approx$
	LLM-MTPSO	0.91(0.02)	—
5	SelfpacedEnsemble	0.51(0.01)	$\approx$
	BalanceCascade	0.50(0.01)	$\approx$
	SMOTEBoost	0.42(0.01)	—
	AsymBoost	0.38(0.01)	—
	LLM-MTPSO	0.51(0.01)	—
6	SelfpacedEnsemble	0.48(0.01)	$\approx$
	BalanceCascade	0.48(0.01)	$\approx$
	SMOTEBoost	0.32(0.01)	—
	AsymBoost	0.30(0.01)	—
	LLM-MTPSO	0.49(0.01)	—
7	SelfpacedEnsemble	0.44(0.01)	—
	BalanceCascade	0.44(0.01)	—
	SMOTEBoost	0.31(0.01)	—
	AsymBoost	0.26(0.01)	—
	LLM-MTPSO	0.46(0.01)	—
8	SelfpacedEnsemble	0.38(0.01)	—
	BalanceCascade	0.39(0.01)	—
	SMOTEBoost	0.31(0.01)	—
	AsymBoost	0.26(0.01)	—
	LLM-MTPSO	0.43(0.01)	—
9	SelfpacedEnsemble	0.41(0.01)	$\approx$
	BalanceCascade	0.41(0.01)	$\approx$
	SMOTEBoost	0.27(0.01)	—
	AsymBoost	0.24(0.01)	—
	LLM-MTPSO	0.41(0.01)	—

1–4, LLM-MTPSO consistently outperforms all baselines. For the TE datasets (i.e., datasets 5–9), the overall BAC values are generally lower. Nevertheless, LLM-MTPSO consistently maintains a leading position with competitive results across these challenging benchmarks. Statistical testing results further confirm the significance of the improvements, with most comparisons showing either superiority (—) or statistical equivalence ($\approx$). These observations validate the effectiveness and robustness of LLM-MTPSO in handling imbalanced classification tasks to some extent.

To verify the impact of the LLM component proposed in this paper on the overall algorithm, we conducted an ablation study. As shown in Fig. 2, removing the LLM module results in a performance drop on most datasets, while some datasets remain stable. In contrast, incorporating LLM-assisted fusion generally improves the BAC, especially on datasets with

higher classification difficulty. These results demonstrate that the LLM component contributes positively to performance and plays a supportive role in enhancing the generalization ability of the proposed LLM-MTPSO framework.

V. CONCLUSIONS

In this work, we proposed an LLM-assisted multi-task optimization framework to address the challenge of imbalanced classification, especially the difficulty in recognizing minority classes. By decomposing the original multiclass problem into a series of binary subtasks and leveraging LLMs to fuse knowledge across related tasks, our method effectively enhances minority class representation and improves classification performance. Experimental results on four real-world industrial datasets and five TE datasets indicated that the proposed approach achieves a superior performance compared to traditional ensemble methods. Additionally, this framework demonstrates the potential of integrating LLMs as semantic-level knowledge aggregators in evolutionary optimization.

ACKNOWLEDGMENT

This work was supported by grants from the Guangdong Science and Technology Programme 2024B0101120003, the Natural Science Foundation China 72271168, the Shenzhen Science and Technology Program KJZD20230923114111021, the Guangdong Basic and Applied Basic Research Foundation 2024A1515012485, and the Shenzhen Fundamental Research Program JCYJ20220810112354002.

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